10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 7.3 — Trigonometric Problems in Three Dimensions and in Real Life

Unit 7 | Solved Exercise | New Book (PECTAA)

Exercise 7.3 (Solved)

In a three-dimensional problem the angle between a line and a plane is the angle between the line and its projection on that plane. First draw the right triangle in the plane, find the two lengths, then use tan θ = opposite ÷ base.

1. In the triangular prism, find (i) the length CF, (ii) the length BF, (iii) the angle BFC, correct to one decimal place.

From the figure: FE = 5 cm, CE = 4 cm, BC = 3 cm. The face FECD is a rectangle, so CE is perpendicular to FE, and BC is perpendicular to this face.

(i) the length CF

In the right-angled triangle FCE

$CF^2 = FE^2 + CE^2$

$CF^2 = 5^2 + 4^2 = 25 + 16 = 41$

$CF = \sqrt{41} = 6.40$ cm

Answer: CF ≈ 6.4 cm

(ii) the length BF

BC is perpendicular to the face FECD, so BC ⊥ CF

In the right-angled triangle BCF

$BF^2 = BC^2 + CF^2$

$BF^2 = 3^2 + 41 = 9 + 41 = 50$

$BF = \sqrt{50} = 7.07$ cm

Answer: BF ≈ 7.1 cm

(iii) the angle BFC

$\tan\theta = \frac{BC}{CF} = \frac{3}{6.40} = 0.4686$

$\theta = 25.1°$

Answer: ∠BFC ≈ 25.1°

2. The diagram shows a triangular pyramid with a horizontal rectangular base PQRS, in which PQ = 16 cm and QR = 10 cm. M is the midpoint of PR. The vertex T is vertically above M and MT = 15 cm. Calculate the size of the angle between TP and the base PQRS. Give your answer correct to 1 decimal place.

$PR^2 = PQ^2 + QR^2 = 16^2 + 10^2 = 256 + 100 = 356$

$PR = \sqrt{356} = 18.87$ cm

$PM = \frac{PR}{2} = \frac{18.87}{2} = 9.43$ cm

studysapiens.com

MT is perpendicular to the base, so PM is the projection of TP

$\tan\theta = \frac{MT}{PM} = \frac{15}{9.43} = 1.591$

$\theta = 57.8°$

Answer: Angle between TP and the base ≈ 57.8°

3. The diagram shows a pyramid. The base PQRS is a horizontal square of side 10 cm. The vertex V is vertically above the midpoint M and VM = 12 cm. Calculate the size of angle VPM.

$PR^2 = 10^2 + 10^2 = 200$

$PR = \sqrt{200} = 14.14$ cm

$PM = \frac{14.14}{2} = 7.07$ cm

VM is perpendicular to the base, so PM is the projection of VP

$\tan(\angle VPM) = \frac{VM}{PM} = \frac{12}{7.07} = 1.697$

$\angle VPM = 59.5°$

Answer: ∠VPM ≈ 59.5°

4. ABCDE is a square based pyramid, in which AE = BE = CE = DE = 12 cm and AB = 15 cm. Calculate the size of angle DEB. Give your answer in degree (whole numbers).

$BD^2 = AB^2 + AD^2 = 15^2 + 15^2 = 225 + 225 = 450$

$BD = \sqrt{450} = 21.21$ cm

In triangle DEB: DE = EB = 12 cm, DB = 21.21 cm

$\cos\theta = \frac{DE^2 + EB^2 - DB^2}{2 \times DE \times EB}$

$\cos\theta = \frac{144 + 144 - 450}{2 \times 12 \times 12} = \frac{-162}{288} = -0.5625$

$\theta = 124.2°$

Answer: ∠DEB ≈ 124°

5. The diagram shows a triangular prism with a horizontal rectangular base ABCD. AB = 10 cm, BC = 7 cm, M is the midpoint of AD. The vertex T is vertically above M and MT = 6 cm. Calculate the size of the angle between TB and the base.

AD = BC = 7 cm, so AM = 3.5 cm

$BM^2 = AB^2 + AM^2 = 10^2 + 3.5^2 = 100 + 12.25 = 112.25$

$BM = \sqrt{112.25} = 10.59$ cm

MT is perpendicular to the base, so BM is the projection of TB

$\tan\theta = \frac{MT}{BM} = \frac{6}{10.59} = 0.5666$

$\theta = 29.5°$

Answer: Angle between TB and the base ≈ 29.5°

6. In an isometric game, the camera is placed at a 45° angle from the ground. If the player is 10 units in front and 10 units above, what is the direct line of sight distance?

Horizontal distance = 10 units, vertical height = 10 units

studysapiens.com

Distance^2 = 10^2 + 10^2 = 100 + 100 = 200

Distance = √200 = 14.14 units

The angle 45° confirms tan 45° = $\frac{10}{10}$ = 1

Answer: Direct line of sight ≈ 14.1 units

7. A surveyor spots the top of a tower at an elevation angle of 28°. He is standing 60 metres from the base. Find the height of the tower.

tan 28° = $\frac{h}{60}$

h = 60 × tan 28° = 60 × 0.5317

h = 31.9 m

Answer: Height of the tower ≈ 31.9 m

8. A boat crosses a river 80 m wide, at a 60° angle to the current. What distance does the boat actually travel?

The width of the river is the side opposite to 60° in the right triangle, and the path of the boat is the hypotenuse.

sin 60° = $\frac{80}{d}$

d = $\frac{80}{\sin 60°}$ = $\frac{80}{0.8660}$

d = 92.38 m

Answer: Distance travelled ≈ 92.4 m

9. A listener hears a sound from two speakers. One is 6 m directly ahead and the other is at 30° to the side, 6 m away. Find the distance between the speakers.

Both distances are 6 m and the angle between them is 30°

$d^2 = 6^2 + 6^2 - 2(6)(6) \cos 30°$

$d^2 = 36 + 36 - 72(0.8660) = 72 - 62.35$

$d^2 = 9.65$

d = 3.11 m

Answer: Distance between the speakers ≈ 3.1 m

10. A 10 m ladder leans against a wall making an angle of 75° with the ground. How high does it reach up the wall?

h = 10 sin 75°

h = 10 × 0.9659

h = 9.66 m

Answer: Height on the wall ≈ 9.66 m

studysapiens.com

11. A lighthouse is located on a cliff 80 m above sea level. A ship is spotted at an angle of depression of 15°. How far is the ship from the base of the cliff?

The angle of depression from the top equals the angle of elevation from the ship, so the right triangle has 15° at the ship.

$$\tan 15° = \frac{80}{d}$$

$$d = \frac{80}{\tan 15°} = \frac{80}{0.2679}$$

$$d = 298.6 \text{ m}$$

Answer: Distance from the base ≈ 298.6 m

Remember — Angles in Three Dimensions

Angle between a line and a plane = angle between the line and its projection on the plane.

Draw the right triangle inside the plane first, then use tan θ = opposite ÷ base.

Angle of elevation is measured upward from the horizontal; angle of depression is measured downward, and both are equal for the same two points.

For a pyramid standing on its base, the foot of the height is the centre of the base.

studysapiens